

XINGRONG DIAO

✉ 12332163@mail.sustech.edu.cn • ☎ +86 15889649011 • 🏠 [Homepage](#) • 🎓 [Google Scholar](#)

RESEARCH INTERESTS

Deep reinforcement learning, embodied intelligence, map-free multi-agent navigation, contact-rich manipulation, and humanoid robot learning.

EDUCATION

Southern University of Science and Technology • Shenzhen, China 2023.09 – Present

M.Eng. in Electronic Science and Technology | GPA: 3.59/4.0

Thesis: *Research on Robot Navigation and Manipulation Algorithms for Unstructured Scenarios*

Research Focus: Deep reinforcement learning, Embodied intelligence, map-free navigation, and contact-rich manipulation

Southern University of Science and Technology • Shenzhen, China 2019.09 – 2023.07

B.Eng. Robotics Engineering | GPA: 3.73/4.0

Core Courses: Advanced Artificial Intelligence, Modern Signal Processing, Robot Modeling & Control

RESEARCH EXPERIENCE

Humanoid Robot Learning and Whole-Body Interaction 2024.08 – Present

- Investigated **humanoid robot behavior learning and whole-body interaction** with reinforcement learning, motion imitation, and simulation-to-real transfer for locomotion and contact-rich tasks.
- Developed **Contact-Track**, a task-conditioned interaction framework that represents interaction objectives through contact specifications (contact state, contact position, contact direction, object pose, and target body parts) instead of predefined full-body references.
- Conducted **sim-to-real deployment of humanoid Mimic policies**, including URDF modification, kinematic calibration, and real-world deployment.

LiDAR-Based Multi-Agent Navigation 2024.02 – Present

- Designed **LSTP-Nav**, a lightweight map-free multi-agent navigation framework that directly maps raw 2D LiDAR observations to motion commands for decentralized collision avoidance.
- Proposed **LSTP-Net**, a compact spatiotemporal policy network integrating GRU-based temporal modeling and attention-based feature aggregation to address partial observability and dynamic-agent interactions with low computational overhead.
- Introduced a heading-stability reward and physics-realistic replay simulator, achieving **40 Hz real-time inference** on CPU-only embedded hardware.
- Resulted in **1 first-author journal paper** (published in IEEE TASE) and **1 patent application** under examination.

Contact-Rich Manipulation Planning 2024.02 – Present

- Developed a **cascaded optimization framework for contact-rich manipulation** by decomposing active contact reasoning into Contact Selection Optimization (CSO) and Contact Planning Optimization (CPO).
- Proposed a surrogate contact model and mesh sampling-exchange strategy to solve nonsmooth contact selection as a discrete-continuous optimization problem over object-surface contacts.
- Developed prior-guided contact planning with a ranking strategy to bridge contact priors and online trajectory optimization for complex manipulation tasks.
- Achieved sub-millisecond single-step QP solving within CSO and validated the framework on challenging contact-rich manipulation tasks involving 6D nonprehensile manipulation, tilted pushing, soccer dribbling, and grasping.
- Resulted in **1 first-author paper** under review.

GNN-Guided Motion Planning 2023.07 – 2024.02

- Developed a **GNN-guided sampling-based path planning method** that predicts neighbor guidance values from obstacle representations, searched-path history, and random geometric graph topology to reduce expensive collision checking in high-dimensional motion planning.
- Designed an environment-path encoding architecture integrating GAT/GCN-based graph embedding, 3D-CNN obstacle encoding, RNN path encoding, and attention-based feature fusion for efficient obstacle-aware neighbor selection.
- Validated the method in 7-DoF KUKA simulations and 6-DoF Kinova real-world experiments, reducing collision checks by **99.7%** compared with PRM and planning time by **98.6%** compared with BIT* in R6 tests.
- Resulted in **1 first-author journal paper** and **1 patent application**.

- **Contributed to LR-MPC**, a learning-based risk-aware navigation framework for socially aware robot crowd navigation in dense human-robot shared environments.
- Integrated a Probabilistic Ensemble Neural Network (PENNN) with model predictive control to predict waypoint-level navigation risk while jointly considering efficiency, collision safety, and social-space compliance.
- Designed an uncertainty-aware decision pipeline for robust waypoint selection under dynamic crowd interactions.
- Combined Multi-RRT-based global guidance with real-time MPC execution to support adaptive waypoint selection, improving navigation success rate and social awareness in simulation and real-world experiments.
- Resulted in **1 co-authored journal paper** (published in IEEE TSMC: Systems).

TECHNICAL EXPERTISE

- **Programming:** Python (Advanced) • C++ • MATLAB/Simulink
- **Tools:** ROS • Gazebo • PyBullet • MuJoCo • Isaac Gym • Linux • Git • LaTeX
- **Hardware:** STM32 • Raspberry Pi • Franka • Kuka • Kinova
- **Languages:** Mandarin Chinese (Native) • English (TOEFL iBT: 87)

PUBLICATIONS

Peer-Reviewed Publications

- [J1] **Diao, X.**, Sun, Z., Peng, J., Zhu, B.-K., Jia, B., Wang, J. (2026). LSTP-Nav: Lightweight Spatiotemporal Policy for Map-Free Multi-Agent Navigation With LiDAR. *IEEE Transactions on Automation Science and Engineering*, 23, 14837–14851. [DOI](#)
- [J2] Sun, Z., **Diao, X.**, Wang, Y., Zhu, B.-K., Xu, W., Wang, J. (2026). Socially Aware Robot Crowd Navigation via Online Uncertainty-Driven Risk Adaptation. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 1–14. [IEEE Xplore](#) [DOI](#)
- [J3] **Diao, X.**, Chi, W., Wang, J. (2024). Graph neural network-based method for robot path planning. *Biomimetic Intelligence and Robotics*, 4(1), 100147. [DOI](#)

Preprints / Under Review

- [P1] Zhang, Z.*, **Diao, X.***, Liang, H., Yang, H., Zhu, B.-K., Zhang, D., Wang, J. (2026). Simultaneous Contact Selection and Planning for Contact-Rich Manipulation with Cascaded Optimization. *Under Review*. [arXiv:2605.27972](#)
- [P2] Zhou, R., Song, Q., Ma, Y., Wang, A., Liu, W., Ma, G., Wang, Y., Wang, G., **Diao, X.**, Wang, J., Zhu, C., Liu, X., Ren, H. (2026). SurgLAT: Surgical Latent Attention Tracking for Depth-Aware Robotic Laparoscope Control. *arXiv*: 2608.07876. [arXiv](#)

*Equal contribution.

PATENTS

Primary Inventor

- [1] **Diao, Xingrong**, Ke, Wende. “Underwater mobile robot with adsorption function”, CN219790500U (Granted Utility Model).
- [2] Wang, J., **Diao, X.** “Robot path planning method, electronic device and storage medium”, CN118700132A (Invention).
- [3] Wang, J., **Diao, X.** “A method, device, terminal equipment, and program product for robotic navigation processing”, CN120593752A (Invention; Pending).

AWARDS

- **National First Prize**, Robot Innovation Competition, China Robot & AI Competition (2021)
- **Global Runner-Up**, MATE International Underwater Robot Competition, Explorer Class Telepresence (2022)
- **Graduate Special Scholarship**, SUSTech (2023)
- **National Scholarship**, SUSTech (2025)

LEADERSHIP

- **Treasurer**, SUSTech Shuren Guitar Club – managed budgeting, reimbursement records, and financial coordination for club activities.
- **Performer**, SUSTech 10th Anniversary Gala, 2023 Chinese Traditional Music Gala, and 2022 Freshman Welcome Ceremony.